MILITARY SPECIFICATION
CARTRIDGE, CALIBER .50, SPOTTER-TRACER, M48A1E1

1. SCOPE

1.1 This specification covers Cartridge, Caliber .50, Spotter-Tracer, M48A1E1 for use in Rifle, Spotting, Caliber .50, M8C.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition

DRAWINGS

Munitions Command

DX10542445	- Cartridge, Caliber .50, Spotter-Tracer, M48A1E1
C7643674	- Classification of Cartridge Case Defects
F7553719	- Packing and Marking, Cartridges, Caliber .50, Spotter-Tracer; Cartons; Box, Ammunition, M2A1; Box, Wirebound
IEL-X10542445	- Inspection Equipment List for Cartridges, Caliber .50, Spotter-Tracer, M48A1E1

PUBLICATIONS

TECP 700-700, Vol. III	- Manual of Test Methods for Small Arms Ammunition

FSC 1305

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(Copies of specifications, standards,drawings and drawings and publications required by suppliers in connection with specific procurement functions shold be obtained from the procuring actitity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge shall comply with the requirements specified on Drawing DX10542445, referenced specifications and the folowing:

3.2 Bullet extraction.- The force required to extract the bullet from the cartridge case shall be not less than 200 pounds.

3.3 Bullet sensitivity.The bullet shall not detonate when the nose of the cartridge is caused to impact against a stationary steel plate with a velocity at the point of impact of 55-5 feet per second (ft/sec).

3.4 Residual stress.- The cartridge case shall not split when subjected to a 1.0% mercurous nitrate solution for 15 minutes.

3.5 Waterproof.- The cartridge shall not release more than one bubble of air when subjected to a pressure differential of 7½ pounds per square inch (psi) for 30 seconds.

3.6 Accuracy.- The average of the mean radii of all targets of the sample cartridges fired at 600 yards shall not exceed 5 inches.

3.7 Trace.- When viewed from a line parallel to the plane of the trajectory at each point of observation, the bullet of the cartridge shall exhibit a visible trace from a point not greater than 100 yards from the muzzle of the weapon to a point not less than 1500 yards from the muzzle.

3.8 Impact sensitivity.- The bullet *of* the cartridge, when fired, shall produce a flash and puff of smoke upon impact against a steel plate placed at 175 yards from the muzzle and set at an angle of eleven degrees (11°) with respect to the horizontal

3.9 Velocity.- The average velocity of the sample cartridges, conditioned at 70° ± 2° Fahrenheit (F), shall be 1745 ft/sec plus or minus 20 ft/sec at 78 feet from the muzzle. The standard deviation of the velocities shall not exceed 14 ft/sec.

3.10 Chamber pressure.- The average pressure of the sample cartridges, conditioned at 70 ± 2°F, shall not exceed 38,000 psi.

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3.11 Function and casualty. - The cartridge shall function without casualty.

3.12 Airtight of base closure seal. - The bullet shall not release more than one bubble-of air under the application of a pressure differential of 2 psi.

3.13 Workmanship.- The requirements for workmanship are as specified on applicable drawings, referenced specifications and the following:

3.13.1 Metal defects.- The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.13.2 Foreign matter.- The cartridge shall be free of corrosion, stains, discolorations, dirt, oil and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure suppliers and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 600 cartridges. The sample shall be manufactured using the same materials, equipment processes and procedures as will be used in regular Production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source,the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility specified in the contract.

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4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall results in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification. - Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number and the supplier's identification as assigned by the procuring activity. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shallbe performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The Acceptable Quality Level (AQL) for the major class shall be 0.25 percent and the AQL for the minor class shall be 1.50 percent. All non-conforming cartridges shall be rejected.

4.3.2.1 Classification of defects.- The classification of defects shall be as specified in Table 1.

Table I

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	Cartridge				
1	Discolored, dirty, oily, smeared			X	
2	Corroded; or stained, if etched		X		
	Case				
4	Round head		X		
5	Dent				X
6	Split case				
	in K, L or M location	X			
	in I, S or J location		X		
7	Perforated case	X			
8	Draw scratch				X
9	Scratch			X	
10	Beveled underside of head		X		

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Table I (Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
11	Case mouth not crimped in cannelure		X		
12	Scaly metal				X
13	No chamfer on head (rim)		X		
14	Fold			X	
15	Wrinkle			X	
16	Buckle			X	
17	Bulge			X	
18	Illegible or missing head stamp			X	
19	Defective head			X	
20	Defective mouth			X	
21	No visible evidence of mouth anneal		X		
	Bullet				
22	Dent			X	
23	Scratch			X	
24	Split bullet jacket		X		
25	Loose bullet		X		
26	Missing cannelure		X		
27	Scaly metal				X
28	Upset (crooked) point			X	
31	Defective cannelure			X	
31A	Missing insert in nose		X		
31B	Deformed nose opening		X		
	Primer				
32	No primer	X			
33	Cocked primer	X			
34	Inverted primer	X			
35	Loose primer		X		
36	Nicked or dented primer			X	
37	No waterproofing material (primer pocket joint)			X	
38	Defective crimp			X	
	Gaging				
39	Total length		X		
40	Cartridge profile failure (requiring more than 80 lbs dead weight to insert in profile and alignment gage)		X		
41	Diameter of extractor groove, max.		X		
42	Diameter of extractor groove, min.			X	

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Table I (Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
43	Diameter of head		X		
44	Thickness of head		X		
45	Length to shoulder datum		X		
46	Depth of primer		X		
	Weighing				
47	Weight, min. 2/	X			

1/ Refer to MIL-STD-636 for visual defect standards for defects 1 through 38 with the exception of defects 31A and 31B.

2/ Each lightweight cartridge shall be disassembled and the propellant weighed. Each such cartridge found to contain 100 grains or more of propellant shall be classed as a major defect. Any cartridge containing less than 100 grains of propellant shall be classed as a critical defect.

4.3.2.2 Flash tube. - The debulleted cases obtained from the bullet extraction test shall be examined for "missing flash tube" and "lacquer seal on flash tube missing or punctured." The occurrence of either defect in any cartridge case shall be cause for rejection of the lot.

4.3.3 Tests.- The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.- The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests and shall have been selected in such a reamer that the sample is representative of the entire lot. The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

Table II

Test	Number of Cartridges	Requirement Paragraph
Bullet extraction 1/	25	3.2
Bullet sensitivity 2/	30	3.3
Residual stress (Mercurous nitrate) 1/	50	3.4
Waterproof 3/	20	3.5

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Table II (Cont'd)

Test	Number of Cartridges	Requirement Paragraph
Accuracy 4/	90	3.6
Trace 5/	50	3.7
Impact sensitivity 5/	50	3.8
Velocity 4/	20	3.9
Chamber pressure 4/	20	3.10
Function & casualty 6/		
Rifle, Spotter, Cal. .50, M8C	200	3.11
Airtightness of base closure seal 7/	--	3.12

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartidge fails in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

2/ Failure of one or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. No second sample permittted.

3/ Failure of seven or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than three but less than seven cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if in the combined first and second sample seven or more cartridges fail to comply with the applicable requirement.

4/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of' a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

5/ Failure of 15 or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than five but less than 15 cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. The lot shall be rejected if, in the combined first and second sample, 15 or more cartridges fail to comply with the applicable requirement.

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6/ The lot shall be rejected when function and casualty defects plus all other firing defects exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the first tests exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample, consisting of double the quantities specified under function and casualty test, shall be fired.If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

7/ The bullets extracted from the cartridges in the bullet extraction test shall be used for this test. Failure of seven or more bullets to comply with the applicable requirement shall be cause for rejection of the lot. If more than three but less than seven bullets fail in the first test, a second sample consisting of double the number of bullets in the first sample shall be tested. The lot shall be rejected if in the combined first and second sample, seven or more bullets fail to comply with the applicable requirement.

4.3.3.2 Firing defects.- Firing defects and acceptance numbers shall be as specified in Table III.

Table III

Defects	Acceptance First Sample	Acceptance Cumulative (1st & 2nd Sample)
1. Misfire	1	2
2. Bullet remaining in bore 1/	0	-
3. Primer leak		
a. Perforation in firing pin indent in primer cup	1	2
b. Escape of gas through primer cup other than 3a	1	2
c. Escape of gas around primer cup more than 50% of periphery	4	6
d. Blown primer or primer falls out of pocket on retraction of bolt	0	1
e. Primer remains in pocket but is physically loose	2	3
4. Case casualties		
a. Longitudinal split 2/		
(1) Neck and shoulder (I or S)	11	18

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Table III (Cont'd)

Defects	First Sample	Cumulative (1st & 2nd Sample)
(2) Body (J)	2	3
(3) Body (K)	1	2
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential rupture 2/		
(1) Partial, shoulder or body (J&S)	2	3
(2) Partial, body (K)	0	1
(3) Partial, head (L)	0	1
(4) Complete	0	1
c. Flash tube		
(1) Missing flash tube 3/	0	1
(2) Flash tube remaining in bore 1/	0	-
5. Failure to extract	0	1
6. Weapons stoppage 4/	0	1
7. Bursting bullet	0	1

1/ No second sample permitted. Lot shall be rejected.

2/ For location of defects indicated by letters in parentheses, see Drawing C7643674.

3/ A loose flash tube shall carry no penalty.

4/ All stoppages attributable to the ammuntion, with the exception of misfire, complete rupture or failure to extract, observed in all tests shall be included.

4.3.4 Packaging, packing and marking inspection.- During or immediately prior to the packaging operation, 100% examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a high pressure test cartridge shall be classed as a critical defect. Occurrence of any type other than high pressure test shall be classed as a major defect. All nonconforming cartridges shall be rejected. Inspection for packaging, packing, and marking shall be in accordance with MIL-STD-644, as applicable to the drawing.

4.3.5 Inspection equipment.- The examination and tests shall be made using the equipment listed on IEL-X10542445.

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4.4 Test methods and procedures.

4.4.1 Bullet extraction.- The cartridges shall be tested in an approved bullet extractor machine. The rate of travel of the test head shall be not less than three nor more than six inches per minute.

4.4.2 Bullet sensititity.- Each cartridge of the test sample shall be dropped from a height of 50 feet through a one inch nominal diameter tube in such a manner that the tip of the bullet shall strike normal on a flat rigid steel plate.

4.4.2.1 Alternate method.- The alternate method shown in the bullet sensitivity section of the IEL may be used. The bullets shall be extracted from the cartridge cases and inserted into the bullet holder. The pressure of the air gun used shall be adjusted so that the bullet holder assembly is propelled at a velocity of 55-5 feet per second, at the point of striking normal on a flat rigid steel plate.

4.4.3 Residual Strress (Mercurous nitrate). The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.4 Waterproof.- The cartridges (not exceeding five at a time) shall be placed horizontally under a 2 to 2½ inch head of freshly boiled water in a glass container and the container shall be evacuated to a pressure of 7½ pounds per square inch (15 inches of mercury) below atmospheric pressure and held at that pressure for 30 seconds. The number of bubbles liberated from the mouth or primer or both of each cartridge. shall be observed.

4.4.5 Accuracy.- The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.6 Trace.- The test shall be conducted in accordance with the methods and proceedures of TECP 700-700, Vol. III, and the following: Observation for trace performance shall be conducted on a range of at least 1500 yards length. All trace testing shall be performed at night. Observation for trace per-fomance shall be made at the weapon and at points 100 yards and 1500 yards beyond the muzzle of the gun on a line parallel to and approximately 75 yards from the plane of the trajectory. Lights or other suitable visible markers shall be placed opposite the 100 and 1500 yard observing points to insure a line of sight from each point perpendicular to the plane of the trajectory. The ammunition shall not be conditioned to temperature but shall be fired under existing conditions. Three cartridges shall be fired to sight, warm and foul each weapon prior to the test. The sample cartridges shall bebfired in regular sequence permitting sufficient time between shots for each observer to record trace characteristics at each observation point. Each cartridge shall be considered as a defective once only in the event if fails at more than one observation point. Observations for trace performance shall be as follows:

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a. Observation at weapon position. - The observer at this point shall observe for any abnormal behavior as an informational check only. Record shall be made of erratic flight, muzzle flash of obviously greater intensity than that produced by other bullets in the sample or such other behavior as may be considered abnormal by the observer.

b. Observation at 100 yard position.- The observer, with line of sight established perpendicular to the trajectory by the light or visible marker, shall note the trace performance of the bullet from each sample cartridge. tridge. A trace which is visible prior to 100 yards range shall be considered acceptable. The cartridge shall be considered defective when the bullet exhibits no trace.

c. Observation at the 1500 yard position.- The observer, with line of sight established perpendicular to the trajectory by the light or visible marker, shall note the trace performance of the bullet from each sample cartridge. A visible trace extending to 1500 yards range or beyond shall be considered acceptable. The cartridge shall be considered defective if the bullet exhibits no trace.

4.4.7 Impact sensititity.- The test shall be conducted in accordance with the methods and procedures of the accuracy test in TECP 700-700, Vol. III using the equipment listed for the impact sensitivity test in IEL-X1054245.

4.4.8 Velocity.- The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.9 Chamber pressure.- The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.10 Function and casualty.- The test shall be conducted in accordance with TECP 700-700, Vol. III. Two rifles shall be used and 100 cartridges shall be fired in each. shall be fired in The test shall. be conducted on a range which is a minimum of 100 yards in length.

4.4.11 Airtightness of base closure seal.- The bullets shall be placed base up under approximately 1.5 inch head of freshly boiled water in a vacuum jar test unit. The desiccator or jar shall be evacuated to a pressure of 2 pounds per square inch (4 inches of mercury) below atmospheric pressure and held at that pressure for not less than 5 seconds. The number of bubbles liberated from each bullet shall be observed.

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4.4.12 Defect penalty. - In any ballistic test, except function and casualty, in which the occurrence of a firing defect listed in Table III prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized but the total ballistic sample shall be penalized for such defects in accordance with Table III.

5. PREPARATION FOR DELIVERY

5.1 Packing. Level A (Worldwide shipment).- The cartridges shall be packed in accordance with Drawing F7553719.

5.2 Marking and labeling.- Packing boxes shall be marked and labeled in accordance with the drawing cited in 5.1.

6. NOTES

6.1 Ordering data. - Invitations for bids or request for proposal and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provisions for the supply, maintenance and disposition of mandatory ballistic test equuipment for acceptance inspection purposes.

6.1.4 provisions for the submission of acceptance of inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.1.5 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

6.2 Safety precautions. - Fumes produced by functioning of the spotter-tracer bullet are considered toxic and inhalation of such fumes should be avoided. Adequate exhausts should be provided to remove fumes from test areas.

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6.3 International agreements.- Certain provisions of this specification are the subject of ABC-Standard- 46A, dated 4 March 1958; Amendment No.1, dated 10 June 1959 and ABCA-Amy-Standard-157, dated 23 September 1966. The international agreements concerned will be considered in formulating any proposal for amendment, revision or cancellation of this specification. When departmental custodians propose to make any changes which might affect compatibility of the item with international standardization agreements, they will so inform their respective Departmental Standardization Office so that appropriate action may be taken respecting the international agreements concerned.

Custodian:

Army - AR

International interest (See section 6)

Preparing activity:

Army - AR

Project No. 1305-A516

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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